

This is the Accepted Manuscript version of an article published by Elsevier in the *Habitat International* in 2024, which is available at: <https://doi.org/10.1016/j.habitatint.2025.103594>
Cite as: **Doan, Q. C.**, Zhang, X.* (2025). The costs of delay: Marginalization and sustainability challenges around Hanoi's urban rail projects. *Habitat International*, 166, 103594. <https://doi.org/10.1016/j.habitatint.2025.103594>

The costs of delay: Marginalization and sustainable development challenges around Hanoi's urban rail projects, Vietnam

Quang Cuong Doan ^a, Xiaohu Zhang ^{a b *}

^aDepartment of Urban Planning and Design, Faculty of Architecture, The University of Hong Kong.

^b Urban Systems Institute, The University of Hong Kong, Hong Kong.

Abstract: Existing studies on urban rail and transit-oriented development tend to emphasize their positive impacts on urban life, such as enhanced accessibility and increased real estate value. However, it remains unclear how prolonged delays in implementing urban rail projects affect communities, particularly under a sustainable development lens. This study offers a detailed analysis of the under-researched social issue of marginalization by drawing on the concepts of megaproject social responsibility and state-led infrastructure development. Our empirical data are derived from an urban rail line in Hanoi, Vietnam, which experienced prolonged construction and operational delays. Using an in-depth interview method, we highlight the economic, social, and environmental hardships endured by affected residents during construction and delayed periods. The analysis reveals that households living near the rail stations have suffered substantial losses in income, deteriorating accessibility, and declining environmental quality due to prolonged construction activities. These adverse outcomes are attributed to infrastructure development strategies that have not prioritized citizen well-being. Despite ongoing public complaints, the persistent negative impacts of the rail projects remain unresolved, mainly due to the weak management and cooperation of the state.

By examining the case of Hanoi's urban rail line, this paper provides an analytical framework to better understand the social consequences of urban development processes in Vietnam and, more broadly, in Global South contexts.

Keywords: Urban rail transit; Marginalization; Social responsibility; Sustainable development; Hanoi

1. Introduction

Urban rail transit (URT) has become an essential part of contemporary urban infrastructure. It is widely recognized for its efficiency, safety, and capacity to alleviate congestion in increasingly dense urban environments. As cities confront rapid urbanization, environmental degradation, and mobility inequality, URT provides a sustainable substitute to private vehicle use, enabling mass mobility while reducing environmental externalities (Ahn et al., 2020). As a result, large metropolises such as Tokyo, Hong Kong, and New York have developed extensive urban rail systems that facilitate millions of daily trips and underpin urban life's functional and geographical structuring (Su et al., 2024).

A growing body of scholarship examines the impacts of urban projects on communities. On the one hand, researchers have identified the significant benefits of URT projects for communities. From an economic standpoint, numerous studies indicate that proximity to metro stations enhances residential desirability, increasing housing prices within station catchment areas (Ahn et al., 2020; Doan, 2023; Doan et al., 2024). Although the effects of the housing market are well-researched, few studies have explored URT's influence on broader urban economies, such as labor dynamics and business development. Among these, Beyazit (2015) investigated how metro expansion catalyzed structural shifts in local economies, stimulating investment and facilitating a transition from manufacturing to service-based industries in Istanbul. Social effects of URT have received growing attention, particularly in advancing transportation equity and improving public health. By enhancing access to jobs, services, and

public amenities, expanded metro systems improve mobility for traditionally underserved groups, including women, children, and low-income populations (Song et al., 2018; Gachanja and Yang, 2025), and the health of local residents (Chen et al., 2021). On the other hand, scholars also pointed out that the advantages of urban rail development are sometimes exaggerated. For instance, limited metro coverage may exacerbate access disparities in peri-urban areas, particularly healthcare services (Liu et al., 2021). Income-based disparities in public transportation benefits have also been observed, with lower-income groups often deriving the least advantage from investment (Bureau and Glachant, 2011). Improvements in transport infrastructure can also lead to uneven changes in land use, population density, and employment opportunities, with localized effects varying widely by station area and system design (Redding and Turner, 2015). Environmental concerns further reinforce the case for URT expansion. While URT systems typically reduce per capita emissions (Zheng et al., 2019), their net environmental impact remains debated. Some studies challenge the assumption that rail construction and operation automatically lead to net emission reductions, noting that such projects can increase road traffic during construction phases and indirectly stimulate urban expansion and car dependency (Rivers et al., 2020; Ou et al., 2024). These contested outcomes underscore the need to examine URT impacts beyond operational phases, particularly in contexts where governance shapes infrastructure equity.

Despite extensive research on URT's economic, social, and environmental impacts, several gaps need further investigation. First, while most studies focus on the socioeconomic and environmental impact during operation (Guo et al., 2022; An et al., 2024), evidence of unintended negative impacts on local neighborhoods during construction and delayed operation, when promised conveniences have yet to materialize, is rare. Despite well-documented operational benefits, the extended construction phase generates unpredictable burdens, traffic congestion, pollution, and economic losses that have not been

systematically assessed. Particularly, construction-phase disruptions, such as prolonged road closures or land acquisition, can destabilize informal economies (Shih, 2019). These early-stage impacts deserve more scholarly attention, as they may undermine household livelihoods and local business stability before any long-term benefits are realized. Second, a synergistic study across several dimensions is also lacking in most existing studies on rail transportation and neighborhood amenities, which mainly concentrate on one dimension (such as local business activities, employment, or retail density measures during urban rail operations) (Du and Zheng, 2020; Feng et al., 2025). The theoretical foundation is straightforward: As more jobs are located near the station, URT reduces commuting costs between station areas and other sites, thereby expanding the local customer base. According to the economic theory of firm location choice, this results in an increase in the density of local companies (Fischer and Harrington Jr, 1996). Third, evidence of the impact of URT on local communities is limited in the Global South. In dominant narratives, largely informed by empirical evidence from the Global North, where neoliberal paradigms shape urban development, the social implications of URT are often cast in favorable terms (Knowles and Ferbrache, 2016; Tarazona Vento, 2017). However, in many countries in the Global South, where State-led infrastructure planning is prevalent, URT projects may produce adverse social outcomes (Huang, 2021). These projects can serve political objectives and reinforce state power, but often lack public consultation and community engagement. For example, poorly managed compensation schemes and involuntary resettlement have provoked significant tensions between the state and affected populations (Tomabene and Nilsson, 2021). Small businesses, especially in informal sectors, face heightened risks from URT construction (Delmelle, 2021). However, it remains unclear how local economic, social issues, and environmental conditions evolve under the influence of delayed mega projects implemented within the framework of State-led development in the Global South (Lesutis, 2022). Understanding how urban rail projects directly impact

individuals provides valuable insights to reduce negative impacts on local communities (Winch, 2017; Corazza et al., 2023). Finally, equitable land value distribution and accessibility justice among communities have not been examined. Inequitable distribution of benefits may lead to social conflicts, potentially reducing community engagement (Mangioni, 2018; Middeldorp and Le Billon, 2019).

This study seeks to address these gaps by examining community marginalization during the prolonged construction of urban railway projects in Hanoi, Vietnam, where State-led urban development underpins most urban development initiatives. Drawing on the mega project social responsibility concept and State-led infrastructure development, we assess how infrastructure projects affect community well-being, household economic stability, environmental quality, and spatial accessibility during delayed implementation. We also provide a multidimensional understanding of how urban rail construction reshapes urban life in the Global South. It aims to contribute to more equitable and resilient transit planning in rapidly urbanizing regions. Importantly, the study shifts analytical attention from URT's frequently highlighted operational benefits to the often-overlooked social and economic costs borne by communities during extended construction periods.

2. Critical challenges for state-led infrastructure development in the Global South

2.1. Megaproject social responsibility

Megaprojects, defined as large-scale, capital-intensive infrastructure initiatives, have become prominent instruments for shaping urban form, enhancing competitiveness, and driving economic growth, particularly in emerging economies (Flyvbjerg, 2014). As these projects proliferate, they have also attracted growing public and scholarly scrutiny concerning their long-term sustainability and social responsibility.

Sustainable infrastructure planning plays a critical role in achieving broader sustainable development goals, including the United Nations Sustainable Development Goals. The notion

of sustainable infrastructure extends beyond environmentally friendly technologies, including planning, design, construction, and operations that ensure economic viability, environmental protection, and social equity (Inter American Development Bank, 2018). In theory, well-executed megaprojects may contribute to sustainable infrastructure systems by fostering more inclusive, resilient, and efficient urban systems (Zeng et al., 2015). However, the reality is much more complex. Many megaprojects labeled as "sustainable" generate significant socio-economic and environmental externalities, including community displacement, restricted access to services, and long-term disruptions (Corazza et al., 2023). These outcomes are especially pronounced when projects are implemented without adequate consultation, management, and compensation mechanisms (Mangioni, 2018). As a megaproject, URT projects may illustrate this tension clearly. While intended to promote sustainable mobility in the long run, their construction processes often exacerbate socio-spatial inequality in the short term (Beyazit, 2015). Land acquisition and resettlement procedures may disproportionately impact low-income and vulnerable populations, limiting their access to essential services and intensifying perceived injustices (Harms, 2012; Xue et al., 2015). This has prompted scholars to examine the governance challenges and risk factors that can determine whether megaprojects achieve inclusive urban transformation or reinforce existing disparities (Esposito et al., 2021; Wang et al., 2021).

Building on this concern, scholars increasingly advocate for governance models that proactively manage both megaprojects' positive and negative externalities. Effective governance frameworks emphasize inclusive stakeholder engagement, transparent risk mitigation, and equitable benefit distribution (Senaratne et al., 2024; Musonda et al., 2025). Of particular concern is the growing recognition that megaprojects generate more acute social risks than conventional infrastructure initiatives (Liu et al., 2018a). These disparities have

triggered protests, petitions, and wider resistance movements, as documented by the Environmental Justice Atlas (Temper et al., 2015).

The megaproject social responsibility (MSR) concept has gained traction in response to such issues. Introduced by Zeng et al. (2015), MSR advocates for a holistic framework prioritizing social well-being throughout the project lifecycle. Rather than treating social issues as secondary to technical or financial concerns, MSR situates them at the core of infrastructure governance. Subsequent studies have applied the MSR lens to assess the socio-environmental impacts of iconic projects such as the Three Gorges Dam (Liu et al., 2023) and Qinghai-Tibet Railway (Gao and Li, 2022). These studies highlight the need for socially responsive planning processes. Despite growing interest in MSR, a unified set of standards or benchmarks to guide its implementation remains lacking to guide its implementation. Nonetheless, the literature converges on the need to address several key dimensions: local economic, social equity, public health and safety (Ashkanani and Franzoi, 2023), urban form (i.e., accessibility, green spaces) (Xue et al., 2015; Rizzo, 2020), and environmental management (Xue et al., 2015; Wang and Wu, 2019).

2.2. Conceptualizing State-led infrastructure development and marginalization

The resurgence of State-led infrastructure development has received growing attention in urban studies, particularly in contexts where the state assumes a dominant role in shaping spatial development and driving economic growth (Liu and Dixon, 2022). In these settings, governments are not merely regulators but key initiators and executors of large-scale infrastructure projects, planning, financing, and overseeing their implementation as part of ambitious urban development agendas (Zhai et al., 2020). These projects often employ top-down approaches and are positioned as strategic tools to enhance national competitiveness, attract investment, and stimulate growth, particularly in rapidly urbanising economies (Yu et al., 2021). This top-down model of infrastructure governance is observable across diverse

national contexts. From Japan's post-war reconstruction to China's Belt and Road Initiative and India's infrastructure development, large-scale investments in expressways, rail systems, and urban renewal have become deeply intertwined with State-led political and economic agendas (Yu et al., 2021). While imperatives of efficiency and modernization frequently justify such interventions, scholars have increasingly questioned their social implications, particularly the risks of dispossession, exclusion, and marginalization faced by affected communities. For example, He and Qian (2023) explored the social contradictions surrounding State-led urban redevelopment in China, underscoring the displacement of long-term residents and the erosion of community attachment to place. These marginalizing effects of State-led projects were also found to affect low-income groups and other vulnerable populations during urban transitions (Shmaryahu-Yeshurun and Ben-Porat, 2020)

Vietnam provides a compelling case for examining the dual nature of State-led infrastructure development. Since the renovation (Đổi Mới in Vietnamese) reforms in 1986, infrastructure has become a central pillar of Vietnam's socialist-oriented modernization project (World Bank et al., 2016). The State plays a leading role in shaping urban infrastructure, allocating land resources, and managing project implementation, from national expressways to URT systems (Revilla Diez, 2016; Leducq and Scarwell, 2018). Vietnam's infrastructure planning has been marked by highly hierarchical, top-down decision-making structures, with limited transparency and citizen participation (Phuc et al., 2014; Leducq and Scarwell, 2018; Turner et al., 2024). Central and municipal authorities typically determine strategic infrastructure priorities, with funding sourced through public investment and foreign concessional loans (Tien, 2019). Such a governance structure often prioritizes political timelines and technical deliverables over participatory processes and social responsibility. For example, new urban zone projects in Ho Chi Minh City have left profound social and environmental impacts, where many households remained in ruined or fragmented

neighborhoods for extended periods while awaiting implementation (Harms, 2012). Similarly, the state's conversion of agricultural land for urban infrastructure development in Hanoi has posed major challenges for households in maintaining their livelihoods (Nguyen et al., 2016). In other cases, socio-economic and environmental disruptions from State-led infrastructure projects have sparked public protests demanding compensation for associated losses (Chau, 2019).

These examples illustrate that while State-led infrastructure development may deliver long-term public benefits, it often entails various forms of urban marginalisation for local communities. Such marginalisation manifests not only through physical displacement but also via more subtle processes: the erosion of daily life quality, reduced accessibility to essential services, increased vulnerability to environmental stress, heightened economic precarity, and growing spatial inequality. These outcomes are frequently intensified by weak project governance.

2.3. A framework for social responsibility of URT project under State-led infrastructure development

This study integrates the concepts of MSR and State-led infrastructure development to assess the impacts of URT construction on adjacent communities (Figure 1). Building on the MSR framework, this study proposes that the social responsibility of URT projects should extend beyond socio-economic and environmental concerns to explicitly include urban form, a dimension often overlooked but important to understanding spatial transformations. The concept of megaproject social responsibility emphasizes that their impacts must be evaluated holistically, grounded in the core principles of sustainable development (Zeng et al., 2015). Therefore, economic, social, and environmental dimensions are integral parts of the URT impact assessment framework. Previous studies have argued that sustainability must encompass not only environmental, economic, and social aspects, but also the quality and

configuration of the built environment (Lin et al., 2017; Bruzzone et al., 2021). Therefore, we developed the concept of URT's social responsibility across four key dimensions: economic, social, environmental, and urban form. This categorization enables a more comprehensive assessment of the financial, equity, ecological, and spatial implications of large-scale infrastructure projects.

URT projects, often framed as drivers of inclusive growth and economic resilience (Gupta and Vegelin, 2016), frequently involve mass land acquisition and disruptive construction activities undermining household livelihood capital. Small businesses located near construction sites may face decreased customer traffic, lower revenues, or even temporary closures due to access restrictions and commercial displacement (Tornabene and Nilsson, 2021). Compared with traditional projects, mega projects are more prone to generate social risk (Liu et al., 2018a). Drawing on social disruption theory, scholars argue that rapid and large-scale transformations can fracture local routines and erode community cohesion (He and Qian, 2023). Residents with long-term ties to their neighborhoods often experience emotional dislocation, psychological stress, and a loss of social identity due to forced relocation or structural changes in their surroundings (Liu and He, 2025). While URT systems may eventually improve accessibility and mobility justice (Li and Loo, 2016), the construction period can restrict households' access to schools, hospitals, and local businesses, undermining quality of life. The distribution of inequitable benefits of URT also exacerbates social inequity. For example, in Hanoi, Vietnam, where land value capture remains underdeveloped, unequal distributions of land value uplift further deepen perceptions of injustice (Vu, 2017). Residents who surrender land use rights often receive limited financial compensation, while those who retain their property can benefit more significantly from rising property values (Doan, 2023). Such asymmetries can trigger collective action and conflict between communities, local

governments, and developers, underscoring the importance of inclusive planning and compensation strategies (Liu et al., 2018a).

The environmental dimension of this framework emphasizes the need to account for both long-term benefits and short-term construction-related burdens. Although URT systems contribute to greener mobility by reducing reliance on private vehicles and lowering emissions, the construction phase frequently results in elevated dust, noise, and air pollution levels (Lin et al., 2024). Studies have documented significant short-term environmental degradation around metro construction zones, including excessive PM₁₀ levels and noise disturbances (Kumar, 2003; Paschalidou et al., 2016; Vogiatzis and Kouroussis, 2017). Increased congestion, noise, dust, and other pollutants negatively affect nearby residents' physical and mental well-being (Doan et al., 2025). According to environmental justice theory, these effects are not distributed evenly; vulnerable populations, such as the elderly, children, and low-income groups, are more susceptible due to pre-existing health vulnerabilities and limited resources for mitigation (United Nations Development Programme, 2020).

Urban form, the final dimension of this framework, remains an under-theorized but critical aspect of URT social responsibility. URT construction reconfigures the built environment by introducing new station areas, altering street networks, and reshaping urban density patterns (Sung and Eom, 2024). Unlike planned metropolises such as Tokyo or Hong Kong, where land use and transport systems are highly coordinated through formal top-down planning (Edgington, 2019), cities in the Global South such as Hanoi, Jakarta have evolved through a combination of formal and informal settlements (UN-Habitat, 2012; Thinh et al., 2024), leading to urban fragmentation when infrastructure is implemented. For example, the urban morphology in Hanoi is dominated by narrow and deep residential plots, commonly called "tube houses," and a dense network of internal alleyways, which became increasingly fragmented following the implementation of major transport infrastructure (Thinh and

Kamalipour, 2022). It was estimated that 75-90% of houses were informally built without permission in Vietnam's cities between 1990 and 2000 (UN-Habitat, 2014). Additionally, urban villages have become increasingly popular due to the rapid urban expansion. Such villages often have underdeveloped infrastructure and facilities, and are considered informal or unregulated settlements (Liu and He, 2010). However, these informal spatial systems are shaped by a hybrid of socialist planning, historical legacies, and informal market-driven growth, often falling outside the logic of formal transport planning, which complicates infrastructure integration. In many cases, State-led projects are designed primarily to focus on technical aspects, while neglecting the daily spatial practices of informal settlements (Yip and Tran, 2016). The result is often fragmented urban morphology, where disrupted neighborhoods are replaced by disjointed landscapes. For example, the Ring Road project led by the state cut through informal settlements in Hanoi has produced irregular spatial forms, including extremely small, substandard housing and visibly deteriorated urban environments (Lao dong, 2021). Consequently, improvements in transport infrastructure can lead to uneven changes in urban form, with localized effects varying widely by station area (Turner et al., 2024).

This evidence reflects a more profound governance logic inherent to state-led infrastructure regimes, where social responsibility principles can be ignored. As a result, the planning process prioritizes infrastructure delivery over inclusive, participatory urban transformation. This disjuncture leads to a neglect of community needs in shaping both the process and the spatial outcomes of development. We argue that a socially responsible approach to URT megaprojects must go beyond environmental assessments or economic projections. It should integrate community well-being, equitable access, spatial justice, and inclusive design. By embedding social responsibility into all four dimensions: economic, social, environmental, and urban form, this framework offers a more holistic lens for evaluating the consequences of state-led infrastructure and guiding more equitable development practices.

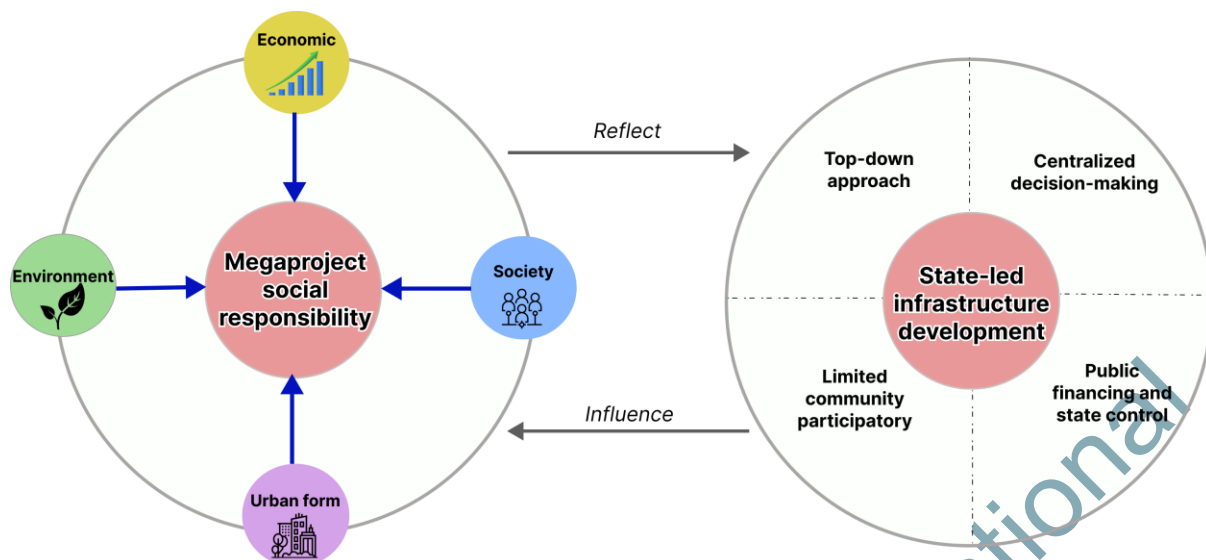


Figure 1. Theoretical framework

3. Research design

3.1. Study area

In this study, we use the urban rail system in Hanoi, the capital of Vietnam, as a case study (see Figure 2). Hanoi is a developing city that covers an area of 3,359.8 km² and has a population of 8.587 million, with an average population density of 2,556 people/km² (General Statistic Office, 2024). In Dong Da central districts (such as those near stations S9, S10, S11, and S12), population density reaches approximately 38,100 people/km² (Hanoi Statistic Office, 2024), surpassing the density of central districts in larger metropolises such as Tokyo. There is a rapid increase in motorbikes, reaching 7 million in 2023 (Hanoi Statistic Office, 2023). With 2.4 million households, the average number of motorbikes per household was approximately 2.9 in 2023 (Hanoi Statistic Office, 2024). Hanoi also faces significant issues with traffic congestion and pollution. In 2018, it ranked as the second most air-polluted city in Southeast Asia (Huu and Ngoc, 2021). These challenges have adversely impacted the economy and quality of life, highlighting the need for local authorities to reduce these harmful effects.

To address severe traffic congestion and promote public transportation, Hanoi has planned a URT system to connect core districts and rural areas (Leducq and Scarwell, 2018).

This urban rail system is considered a mega project in city development, aiming to complete all URT lines by 2030 (The Prime Minister, 2011). Construction of the first two urban rail lines began in 2010 and 2011, with initial completion dates set for 2015 and 2016 (Japan International Cooperation Agency, 2016). However, prolonged conflicts over land clearance compensation with affected residents led to significant construction delays (Asian Development Bank, 2025). As a result, these lines only commenced commercial operations in 2021 and 2024, respectively (Do Duc and Duc, 2025). Studying these delayed URT lines allows us to explore the negative impacts of prolonged construction on residents' lives. Furthermore, as Hanoi is in the early stages of developing its URT network, understanding the adverse effects of construction activities provides valuable insights for local authorities and developers to enhance the quality of life in future projects.

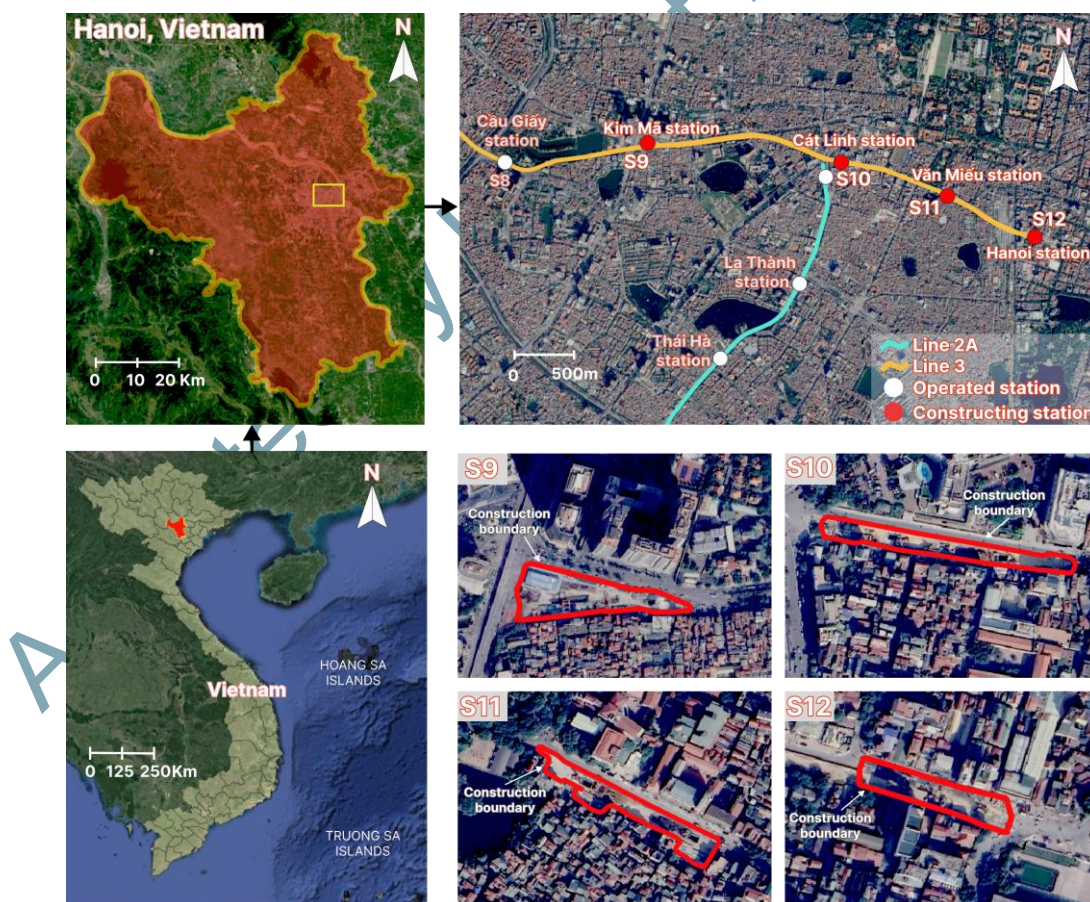


Figure 2. Location of Hanoi, Vietnam and the study areas
Source: Authors' illustration based on OpenStreetMap data and satellite imagery

3.2. Methodology

We conducted a qualitative study to explore the adverse effects of URT construction on local communities. A qualitative approach allows for a focused exploration of human experiences, motivations, and behaviours through a small sample size, facilitating in-depth contextual insights (Sheydayi and Dadashpoor, 2023). Our questionnaire incorporated the principles of sustainable development to comprehensively assess the social, economic, and environmental impacts of urban rail construction. Specifically, we used in-depth interviews to gather qualitative data. We interviewed households in the catchment area of stations S9, S10, S11, and S12 on Hanoi's urban rail line number 3, as these stations experienced long-term construction. The interview questions focused on four main areas: (1) the economic impact of urban rail construction on households, (2) the effects of construction delays on social issues such as employment and job opportunities, (3) the influence of construction on health and environmental aspects, and (4) households' expectations regarding the urban rail project.

Our initial sampling involved convenience sampling based on conversations with households near the construction sites. Snowball sampling was subsequently employed by interviewing households referred by the initial participants. We also contacted the head of a sub-quarter ('tổ trưởng tổ dân phố' in Vietnamese) who has a comprehensive understanding of the study area and residents' expectations. Based on recommendations of the local sub-quarter head, households situated within a 150-meter radius of the construction site experienced the most significant impacts. Accordingly, our study concentrated on households within this recommended buffer zone. This approach allowed us to tap into existing social networks and explore the most pronounced negative impacts on households.

Thirty-four individuals participated in in-depth face-to-face interviews conducted in Vietnamese from July to August 2024. Participation came from a diversity of backgrounds, such as office workers, individual business owners (food, grocery, sidewalk beverage), etc.

(see Appendix 1). Participants ranged in age from 25 to 88 years. Each participant provided consent before the interview began. The interviews, lasting from 15 to 40 minutes, were documented through notes or audio recordings, with participants informed beforehand, and subsequently transcribed, anonymized, and analyzed. Content analysis identified four main themes and 16 subthemes that detail the effects of urban rail construction and delays on residents' lives. Table 1 presents the categorization system.

Table 1. Themes identified through content analysis

Main Category	Sub-category
Economic issues	Income, business, jobs, house and rental market
Social issues	Mobility justice, living conditions, accessibility, compensation price, inequity
Environmental issues	Air pollution, noise pollution, household waste pollution, flood.
Urban form	Urban green, building appearance, road width

4. Result

4.1. Impact of urban rail construction on local economies

We explored how infrastructure development, particularly URT construction, shapes diverse lived experiences. As urban scholars have convincingly argued, development processes shaped by technical experts, are often expected to stimulate local economic activity (Yigitcanlar and Velibeyoglu, 2008; Lan et al., 2023). However, our findings challenge this assumption. Many participants, particularly those operating self-run family businesses near station catchment areas, described a range of detrimental impacts, contrary to expectations of local economic stimulation, including declines in customer traffic and income disruption.

Respondents engaged in individual and household businesses consistently reported a substantial reduction in their customer base, primarily due to decreased accessibility and limited parking availability. Participants described how road closures, redesigns, congestion, and increased travel times discouraged customers from visiting their establishments. While

such disruptions are not uncommon during urban infrastructure projects, respondents emphasized that prolonged construction transformed temporary inconveniences into sustained economic hardship. Reported income declines ranged from 20% to 50%, attributed primarily to decreased customer access during extended construction delays. Many household businesses downsized within six months to a year after construction began. Business owners reported cutting 20% to 30% of their workforce due to declining revenues, and in response, family members sought alternative employment to supplement household income during these challenging periods. Particularly among vulnerable groups such as older people, the construction period severely disrupted income stability. Several respondents emphasized that construction activities had displaced elderly sidewalk vendors, contributing to rising unemployment and deepening precarity. Among them were many who lacked alternative income sources. A restaurant owner in his late 60s expressed his frustrations about the extended disruptions associated with URT construction:

“My family has been running a catering business here for 35 years, but the number of customers has significantly decreased over the past seven years since construction began. Compared to before, we now only have about 40% of our previous customer volume, mostly regulars. Multiple barriers were erected along the roads leading to my shop, making it harder for customers to find us. The access road to my shop is now so narrow that only one motorbike and one pedestrian can pass at a time.” (Interview with interviewee No. 4, July 2024)

This experience is common among many small-scale business households near the stations under construction. Many local households have struggled to maintain their business despite diminished foot traffic. Unable to sustain their enterprises during the protracted construction period, those renting their premises often choose to cease operations or relocate to locations with more viable commercial conditions. Several respondents reported that land

acquisition for the urban rail project had directly displaced their business locations, removing their primary source of livelihood. An 85-year-old woman in a wheelchair shared that she had to stop her beverage business because the business location was reclaimed:

“I had been selling drink vending on Cat Linh Street for nearly 20 years. Although my income was not high, it was enough to support my life as an older person. On weekends, I could earn more than 300,000 VND per day (~12 USD). Since the Cat Linh station (S10) construction started, I no longer have a place to run my business. Selling beverages allowed me to meet people, and chatting with them made me feel happier and healthier.”
(Interview with interviewee No. 21, August 2024)

Most business-owning families agreed that the prolonged construction of urban rail stations has negatively affected their livelihoods across several dimensions. The extended and unpredictable nature of the process eroded livelihood resilience. Although some younger individuals, such as office workers, hold positive views about the eventual positive impacts of URT construction, most older individuals and those working in informal businesses voiced concerns that their livelihoods could not withstand the extended period of disruption. The local rental market for business premises has also declined due to decreased business activities. Before station construction, households could earn 20-30 million VND monthly by renting out their ground-floor spaces. However, few tenants are willing to rent these spaces during the construction period. Even with a 40% reduction in rental prices, occupancy levels remained consistently low during construction. Many families have taken out bank loans to survive crises and sustain their businesses, but they are uncertain how long their business activity can remain viable. These findings highlight how prolonged construction delays amplify local economic strain, especially among individual and household businesses.



Figure 3. Individual business closure (A) and local restaurant with few customers (B)

Source: Authors, July 2024, at Kim Ma (S9) and Cát Linh (S10) stations

4.2. Impact of urban rail construction on social issues

Previous studies have emphasized the positive impacts of urban rail systems on society, such as enhanced employment and recreational activities (Li and Loo, 2016), local amenities (Lin et al., 2024), and improved accessibility (Li et al., 2017). These benefits are typically linked to the operational phase of URT systems. However, our findings indicate that the construction phase, especially when prolonged, can generate significant negative impacts on surrounding communities.

Firstly, approximately two-thirds of interviewees expressed concerns regarding the inconvenience caused by the extensive installation of barriers around construction sites, which occupied a substantial portion of local residential roads. While residents anticipated some disruption, such as restricted accessibility, they did not expect these conditions to persist for more than seven years without a clear timeline for construction completion. Many also reported that the remaining roads had become excessively narrow for the needs of the surrounding neighborhoods. For example, near Văn Miếu station, respondents frequently cited persistent traffic congestion caused by road widths narrowed to 0.8-1.2 meters. Narrow roads forced car-owning households to pay additional monthly fees of approximately 1,500,000 VND (~60

USD) for parking services outside their homes. Although this amount may appear modest, respondents emphasized that, over several years, these cumulative costs represented a substantial financial burden relative to the average annual income of working households. Furthermore, the narrow road conditions raised broader safety concerns. Several respondents recounted near-misses and accidents, including an incident involving a 70-year-old woman who fractured her arm due to a collision in a narrow alley. These examples underscore how temporary inconveniences may escalate into long-term hazards when infrastructure projects experience prolonged delays and inadequate spatial coordination.

"I broke my arm in a traffic accident on this narrow road. When I saw a motorbike approaching, I moved to the side, but my shirt still got caught in the moving vehicle because the road was too narrow. I fell onto the road and was taken to the hospital. My arm has not fully recovered. Even now, I experience fear whenever a motorbike approaches." (Interview with interviewee No. 20, August 2024)

Despite the already constrained road widths, construction workers have further exacerbated the situation by using the space to hang clothes (Figure 4a), reducing road clearance and detracting from the area's aesthetic quality. Similarly, many people expressed concern about a narrow road leading to a primary and secondary school. This 2-meter-wide road serves approximately 1,000 primary students, leading to severe traffic congestion during school arrival and dismissal times. Many parents reported waiting nearly thirty minutes to pick up their children. They expressed frustration that the 10-meter-long road between the primary and secondary schools has been closed due to URT station construction, necessitating an alternate route that adds approximately ten minutes. Their frustration was not only with the inconvenience but also with the absence of a clear end date for the disruption.

Secondly, most metro and URT construction occurs in urban areas, negatively impacting adjacent facilities (Ding et al., 2011; Zhang et al., 2021). Therefore, the planning

and design of URT lines must consider various factors, such as construction techniques and geological conditions (Lin et al., 2024). However, these considerations have been inadequately addressed in the case of URT Line 3. Many households raised concerns that their homes were gradually deteriorating due to geological subsidence. This has heightened concerns about safety within these residents' homes. Numerous houses exhibit large wall cracks, posing direct threats to the occupants' safety and lives. Due to the subsidence, many household items have become unusable. Several families reported shattered windows and doors that can no longer open/close because of deformed door frames. The interviewed households shared that they have reported the situation to the Ward People's Committee and the project developer, but the responses have been unsatisfactory. Project managers have confirmed that the geological subsidence was the cause of the reported structural damage. The Ward People's Committee has visited the affected homes to document the subsidence and wall cracks, dispatched a team to measure the extent of subsidence. Just a few households received support. Households that suffered heavy damage received 5 million VND/month (~USD 196) for renting a house, but this support lasts only 5 months. Less affected households undertook repairs on their houses using their own budget. Despite spending millions of VND on repairs, households reported that structural cracks re-emerged within a short time. A 60-year-old woman whose house has been affected by subsidence and cracks shared.

"I know of many petitions filed by residents in my neighborhood regarding subsidence and cracks in their homes. Some families have reported cracks as wide as 2-3 cm. Currently, I cannot close/open the doors and windows of my house because the ground has subsided. I spent over 2 million VND on repairs, but now the doors are stuck again and cannot be used. I submitted a petition to the local People's Committee; they [People's Committee staff] said they would respond within a week, but it has been over

487 *seven days, and I have not received any responses." (Interview with interviewee No.10,*
488 *July 2024)*

489 Third, all respondents reported that the compensation amounts and support provided
490 were inadequate and unfair. For example, at Văn Miếu station (S11), while the market price of
491 parcels on the main road was around 200 - 250 million VND/m², the compensation rate was
492 approximately 108 million VND/m². Approximately a third of respondents mentioned inequity
493 in compensation. While households that lost 100% of their residential land were relocated to
494 new apartments, those who lost only part of their property were left in unsatisfactory living
495 conditions. Some partially affected households incurred reconstruction costs that exceeded the
496 compensation offered by local authorities. The compensation price was much lower than the
497 market price, which led to a lack of cooperation from the residents. Despite dissatisfaction with
498 compensation rates, some residents ceded their land for URT due to perceived social and moral
499 obligations. However, several residents who joined the opposition campaign were motivated
500 by material interests, such as better compensation and settlement support. During the long
501 opposition, an illustrative case was the campaign for retaining their land until receiving
502 adequate compensation. A male phone store owner expressed dissatisfaction with the
503 compensation fee.

504 *"My family had 4 square meters taken to construct the Van Mieu station. A portion of*
505 *our house had to be dismantled for the station. When part of a house is demolished, it*
506 *affects the entire structure (foundation, ceiling). I only received over 200 million VND in*
507 *compensation (an average of 50 million VND per square meter), but I had to spend more*
508 *than 300 million VND to repair my house. The government's compensation amount does*
509 *not come close to covering the cost of repairs." (Interview with interviewee No.5, July*
510 *2024)*



Figure 4. Narrow roads (A), road connected to schools (B, C) and creaked houses (D, E, F)

Source: Pictures A, B, C, and F were taken by the authors at Văn Miếu and Cát Linh stations (July 2024); pictures D and E from VNExpress.

4.3. Impact of urban rail construction on the environment

URT construction typically comprises three key stages: land acquisition and demolition, construction management, and operation, all of which require detailed management schedules (Ding and Xu, 2017). Construction activities can significantly affect the surrounding environment, including air and noise pollution, flooding, and household waste pollution (Vogiatzis et al., 2018; Islam and Hossain, 2024). While such environmental disruptions are common in large-scale infrastructure projects, their intensity and duration in URT projects have been markedly exacerbated by prolonged construction timelines. This study finds that several environmental disturbances have persisted for over seven years, generating chronic stress among affected communities.

Among the environmental effects, noise and air pollution were the most frequently cited by respondents. Among these, noise pollution received significant attention due to its direct impact on residents' quality of life. Many respondents expressed frustration about the noise that

has persisted since 2018, when construction activities began. Construction work typically starts from early morning until midnight, disrupting residents' sleep. The issue remains unresolved despite numerous complaints to local authorities and the project management committee. Many respondents noted how the noise impacted vulnerable groups, including older people and students. Parents of all five interviewed students reported complaints that construction noise disrupted their children's ability to concentrate during class. They expressed concern about the long-term exposure to such noise. One resident recounted:

"You can hear the noise when they dig the ground for construction. The noise occurs not only during the day but also at night, even when we are sleeping. You cannot imagine how the ground shakes and the loud noises. Previously, I couldn't sleep because of the noise. We eventually installed soundproof doors, but we still get woken up many times by the noise. This is the seventh year since construction started. We hope they finish the project soon." (Interview with interviewee No.06, July 2024)

In addition to noise pollution, respondents expressed frustration over the air pollution they had endured for over seven years. Respondents reported that prolonged exposure to dust and fine particulate matter had significantly disrupted their daily routines. Interviewees stated that keeping doors and windows closed all day and staying indoors was their only viable solution to protect against air pollution from construction activities. Dust pollution was described as particularly severe during excavation and soil transportation, with all participants indicating that they had to clean their homes more frequently to prevent excessive accumulation. One of the primary causes of prolonged dust pollution, as identified by several respondents, was weak project management and a lack of inter-agency coordination. Specifically, different utility providers, such as water, electricity, and telecommunications companies, often failed to synchronize their operations, leading to repeated and unnecessary excavation work. This lack of coordination not only extended construction timelines but also

intensified environmental disruption in surrounding neighborhoods. As one 55-year-old male respondent emphasized, more effective coordination among project teams is essential to minimizing air pollution and reducing the burden on affected communities.

"I believe the parties involved in this project are not coordinating effectively; they seem to be working independently. For instance, the water supply and drainage team dug up the ground to install pipes and then covered it up, and then the telecommunications team dug it up again shortly afterwards to lay cables. This lack of cooperation has led to prolonged and persistent dust pollution." (Interview with interviewee No.32, August 2024)

Furthermore, the majority of respondents criticized the infrequent waste collection, contributing to an unclean environment. Waste trucks collect trash only twice a week because they cannot navigate the narrow roads, exacerbating waste accumulation. Due to limited public awareness, some roads have turned into makeshift household waste collection sites despite littering bans (Figure 5). A small number of respondents also reported occasional flooding during heavy rains. In stations with incomplete drainage systems, standing water and uncollected garbage have become significant sources of pollution, negatively affecting the environment and residents' quality of life. While these issues might have been temporary had construction proceeded on schedule, the prolonged timeline has transformed them into persistent environmental hazards that jeopardize public health and diminish overall quality of life in nearby communities.



Figure 5. Flooding (A), household waste pollution (B), brick on the road (C)

Source: Pictures taken by the authors at Văn Miếu station (S11) and Cát Linh station (S10), August 2024.

4.4. Disquiet over urban form

Many respondents lamented the replacement of residential buildings by URT stations. Numerous houses along the route of Line 3 have been demolished or rebuilt, resulting in smaller living spaces. Some remaining properties have become so narrow, often referred to as ‘super-thin houses’ (nhà siêu mỏng in Vietnamese), that they are deemed unsuitable for habitation. As a result, these properties are frequently abandoned.

Several respondents expressed dissatisfaction with the local authorities' statements about super-thin houses. The authorities frequently criticize these super-thin and irregularly shaped houses, but they mostly result from land acquisition for construction projects, including railway stations. Many families are left with only 5 or 10 square meters of land after land acquisition for the railway stations, with little alternative but to construct narrow dwellings rather than leave the land vacant.

'Why don't they acquire all households within a 400-meter radius of the station? They could use that land for auctions and build properties to sell, generating additional revenue for other transportation projects, as China has done. This way, the authorities could generate additional financial resources and address the issue of unsightly super-thin houses.' (Interview with interviewee No. 04, July 2024).

Furthermore, the remaining demolished materials were inadequately cleared, leaving an unattractive environment and negatively affecting the urban landscape. Even after seven years of construction, the surrounding areas appear disheveled and dirty. This situation starkly contrasts with the stated goal of 'ngõ phố xanh sạch' (spotless, green streets) (Figures 6.B-2024, C-2024). One-third of the interviewees expressed concern over the number of trees cut down for construction. This activity, together with the development of railway infrastructure, has significantly transformed the spatial and ecological character of the old quarter. A 65-year-old restaurant owner, born in Hanoi, detailed the negative impact of URT construction on the urban form:

"Although there weren't many ancient trees along this road before, they were sufficient to alleviate the oppressive summer heat. Those trees were a source of oxygen in this densely populated area. I had hoped they would preserve the trees so people could continue to walk and exercise here. The railway might bring positive changes once completed, but it has been under construction for seven years. I am now 65 years old and still waiting to witness the new urban landscape" (Interview with interviewee No.18, July 28, 2024)

In addition, most respondents expected local authorities and project managers to provide wider roads to improve accessibility. Many roads have been barricaded for construction, leaving only 1 to 1.5 meters of usable space between residential areas and construction zones. This has caused significant inconvenience in daily life. A woman in her 50s, living near S10 station, expressed her concerns:

"You see, the road is just 1.2 meters wide, barely enough for one motorbike and a pedestrian. We feel very anxious walking on this road because the risk of accidents is very high. Moreover, a fire truck cannot reach the site if a fire breaks out, and an

ambulance cannot pass through such a narrow road in emergencies.” (Interview with interviewee No.26, August 2024)



Figure 6. Urban appearance after a long-term URT construction

Source: Picture A was provided by the interviewee, Pictures B and C were taken by the authors (at Văn Miếu station - August 2024)

5. Discussion

URT systems are essential for urban development, reducing personal vehicles, and improving air quality (Chen and Whalley, 2012; Liu et al., 2018b; Zheng et al., 2019). Existing studies have mostly emphasized the positive aspects of URT on the economy and environment at a macro scale (Sun et al., 2015; Chen et al., 2019; Kaneko et al., 2019; Ou et al., 2024). Few studies have explored the negative impacts of URT, and those that do mostly rely on quantitative or secondary data, providing limited insight into how people react to URT projects and how their lives are affected by transit in the urbanization process (Vogiatzis and Kouroussis, 2017; Turner et al., 2024). Using a qualitative method, this study fills the gaps above through the lens of megaproject social responsibility.

Our findings revealed that URT negatively influences the local economy during construction, particularly regarding employment, income, business, and the rental market. We observed that unlike younger age groups who are still in the labor force and can rent alternative

commercial spaces in better business locations, the elderly typically do not have the same flexibility or mobility to adapt their livelihood strategies. Such elderly individuals operate small-scale businesses from or near their homes, such as tea stalls, small eateries, or food vendors, often using sidewalks or other informal spaces that do not require large capital investments (Dang et al., 2025). When urban rail stations enter the construction phase, these spaces are either inaccessible or reclaimed, directly undermining their business operations. Therefore, they are particularly vulnerable to external shocks, making it difficult for them to secure stable livelihoods (Phuong and Sukontamarn, 2024). These findings align with literature highlighting the disproportionate sensitivity of vulnerable groups to infrastructure megaprojects (Orueta and Fainstein, 2008; Sankaran et al., 2022). Urban rail construction has created physical barriers (e.g., blocked sidewalks and increased traffic) that hinder mobility, making it harder for elderly individuals to run their businesses. Unlike larger firms that may possess financial buffers or relocation options, small businesses often suffer losses (Ray, 2015; Tornabene and Nilsson, 2021).

In contrast to Chinese megaprojects, where displaced populations are often compensated through alternative real estate or urban amenities (Wang and Wu, 2019), Hanoi's rail projects have disproportionately focused on land acquisition without providing adequate support to affected households. Despite years of persistent complaints and petitions from residents, the adverse impacts remain largely unaddressed. In fact, land-related grievances have accounted for up to 61.5% of all complaints in recent years (Van Dang, 2022). This raises concerns about the limits of State-led infrastructure development, in which centralized planning suppresses public input and overlooks the social costs. Our case study suggests that such governance approaches may not enhance the efficiency of urbanization but instead intensify local economic harm and generate long-term social conflict.

Our qualitative research findings provide additional evidence to help explain empirical quantitative studies reporting a negative association between housing sales/rentals and urban rail projects. For instance, Zhu and Diao (2022) found that the price premium decreased by 21.4% during the construction phase; however, the underlying causes of this decline were unclear. Our interviews indicate that renters, particularly those dependent on public transit, prefer locations with operational lines, avoiding areas plagued by construction delays. Prolonged construction phases bring ongoing noise, dust, traffic disruptions, and limited access to nearby areas, making living near construction sites less attractive for potential renters (Zhang and Jiao, 2019). Long-term construction can deter renters who prefer quieter, more accessible neighbourhoods (Chasco and Gallo, 2013; Morano et al., 2021). These inconveniences drive current tenants away and deter new renters, leading to higher vacancy rates and declining rents. As landlords struggle to attract tenants, the pressure to reduce rental prices increases, resulting in a suppressed housing market in construction-impacted zones.

In addition to economic losses, our study highlights significant deterioration in residents' quality of life near construction zones. Road closures, lane reductions, and frequent rerouting have disrupted daily commutes and increased traffic congestion (Bhutani et al., 2016). Construction involves heavy machinery, drilling, and other loud activities that often run day and night, disrupting sleep, concentration, and overall well-being, leading to stress and sleep deprivation for residents nearby (Lin et al., 2024). For vulnerable populations, such as the elderly and children, high noise levels can exacerbate health issues, including hypertension and heart disease. These adverse outcomes contrast sharply with practices in many Global North cities, where stronger welfare protections and urban governance frameworks help mitigate the negative impacts of megaproject implementation (Wang and Wu, 2019). In cities such as London, Amsterdam, and New York, regulatory oversight and public-private partnerships often deliver associated public benefits, such as affordable housing, cultural

infrastructure, and inclusive employment, alongside the infrastructure itself (Fainstein, 2008). Such safeguards are frequently absent in Global South contexts like Hanoi, where large-scale urban projects are more likely to proceed with limited attention to the socio-spatial vulnerabilities (Gillespie and Nguyen, 2018).

While some scholars characterize China's megaprojects as a form of 'state entrepreneurialism' that mitigates public backlash through compensation (Kan, 2019; Wang and Wu, 2019; Zhang et al., 2020), we argue that Vietnam's approach is further straining the relationship between the government and the people due to a lack of capacity in project management, planning and coordination. The Vietnamese state employs ambiguous policy narratives and symbolic participatory practices, sometimes using coercion, while largely ignoring the substantive concerns of affected citizens (T.T. Duong et al., 2020; Le et al., 2022). We found that inequity in land value uplift distribution has become more severe since the construction of urban rail lines. In the absence of control policies, as seen in Hanoi, Vietnam, this inequity leads to conflicts between residents and local authorities. Specifically, while landowners outside the acquisition area can attain a price premium without additional taxes, those forced to cede land rights received approximately 108 million VND/m², about half the market price. This inequitable compensation exacerbates social tensions and undermines residents' willingness to support URT projects, often causing delays and eroding trust in local governance (Doan, 2023). Unlike neoliberal megaprojects aimed at capitalizing on land values through private investment (Fainstein, 2008; Watt, 2013), Vietnam's approach is state-driven but lacks meaningful intervention to correct distributive injustices (Thien Thu and Perera, 2011). Although the state has acknowledged some of these disparities, no substantial corrective measures have been implemented.

To minimize the adverse impacts of URT construction in future projects, several key measures should be prioritized. First, we suggest that local authorities implement

socioeconomic stabilization policies for displaced households and those residing near construction sites. These policies should be integrated at the outset of the environmental impact assessment process. Second, project developers must ensure meaningful community participation throughout all stages of URT development. This includes participatory planning processes, transparent communication with affected residents, and the integration of local concerns into project design and implementation (Lin et al., 2017). Third, compensation rates should be adjusted to reflect the market price, ensuring an equitable redistribution. Land value capture mechanisms should also be adopted to finance reinvestment in future urban railway infrastructure. These measures have proven effective in addressing financial constraints and social inequities while mitigating construction delays (Doan, 2023). Fourth, robust environmental mitigation strategies (i.e., noise barriers, dust suppression techniques, and regular waste management) must be enforced to reduce the impact on affected households. In densely populated areas, local authorities may consider temporarily relocating vulnerable residents to reduce health risks associated with prolonged exposure to pollution. Finally, we suggest that Hanoi's URT planning should be aligned with broader urban development strategies, including TOD, preservation of green spaces, and enhancement of pedestrian accessibility. However, achieving these objectives requires strong inter-agency collaboration among transport authorities, urban planners, and land management bodies to ensure a socially responsible and sustainable urban transition.

6. Conclusion

While URT offers numerous socioeconomic benefits, research on the adverse effects of URT on individuals and communities remains limited. This study aligns with the United Nations Sustainable Development Goals, specifically SDG 11, which promotes inclusive, safe, resilient, and sustainable cities. It examines the negative impacts of URT construction and delays on communities from economic, social, environmental, and spatial perspectives. Using

Hanoi, Vietnam, as a case study, we observed that the construction of urban rail has caused significant localized disadvantages for residents across economic, social, and environmental dimensions. The findings indicate that individual business owners and elderly populations are particularly vulnerable to the disruption caused by construction activities, especially regarding income loss. The rental market in areas near construction sites has experienced high vacancy rates, leading landlords to offer discounts of up to 40% to attract tenants. Moreover, environmental conditions have also deteriorated, with increased air and noise pollution levels and diminished basic living conditions. On the social dimension, low compensation rates and inequitable distribution of land value uplift resulting from urban rail construction have generated persistent tensions between the state and affected citizens, leading to negative consequences that eroded residents' willingness to cede land use rights for URT development. The contributions of this study are twofold. Theoretically, it integrates the concept of megaproject social responsibility with State-led infrastructure development to analyse urban rail construction in the context of Vietnam's post-socialist urban regime, revealing how affected communities respond through shifts in livelihood strategies. Empirically, the study introduces a new analytical lens to the existing literature by emphasising the often-overlooked adverse consequences of URT development on adjacent communities.

While our qualitative study offers new insights into the discourse on URT and its impacts on adjacent communities, it has several limitations that should be acknowledged. First, we focused on households and individuals residing within 150 meters of construction stations, even though the effects of URT construction may vary with both distance and project phase. Therefore, future studies could investigate the spatial dimensions of these impacts to provide a more comprehensive understanding of how communities experience such disruptions. Second, we mainly focus on residents' experiences during an ongoing construction phase, meaning that longer-term positive and negative effects remain unknown. Future studies can employ

longitudinal methods to assess how perceptions and outcomes evolve once URT systems become operational. Third, the limited number of empirical studies on URT in Vietnam, particularly those examining its socio-spatial implications, restricts our ability to contrast the Hanoi case with similar infrastructure developments elsewhere in Vietnam.

7. Appendix

Appendix 1. List of interviewees

No.	Gender	Age range	Occupation
1	Female	50-59	Beverage vendor
2	Female	40-49	Office worker
3	Female	20-29	Office worker
4	Male	60-69	Local restaurant owner
5	Male	30-39	Mobile phone shop owner
6	Male	40-49	Office worker
7	Female	50-59	Retired
8	Male	50-59	Local restaurant owner
9	Female	30-39	Food shop owner
10	Female	60-69	Local restaurant owner
11	Male	50-59	Grocery shop owner
12	Female	50-59	Food store owner
13	Female	40-49	Fashion shop owner
14	Female	40-49	Beverage vendor
15	Female	60-69	Appliance store owner
16	Male	30-39	Fashion shop owner
17	Male	60-69	Local restaurant owner
18	Female	60-69	Local restaurant owner
19	Female	50-59	Painting business owner
20	Female	70-79	Retired
21	Female	80-89	Retired
22	Male	60-69	Rice shop owner
23	Female	50-59	Retired
24	Male	50-59	Local motel owner
25	Female	40-49	Former beverage vendor
26	Female	50-59	Housewife
27	Female	40-49	Student's parent
28	Female	40-49	Student's parent
29	Female	40-49	Student's parent
30	Female	30-39	Student's parent
31	Male	30-39	Student's parent
32	Male	50-59	Sanitary equipment shop owner

33	Female	50-59	Fashion shop owner
34	Female	30-39	Fast food shop owner

Appendix 2: Additional photos around stations



Fig. A1. Traffic congestion near Cát Linh station

Source: Authors (August 2024)



Fig. A2. Narrow road at Kim Mã station (A) and Hanoi station (B), and construction activities near Cát Linh secondary school (July 2024)

Source: Authors (August 2024)

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